# ELEG 387V/3124 Systems and Signals
# Course Syllabus

**General Information:**

Instructor: Jingxian Wu
Office: Bell 3168
Email: wuj@uark.edu
Phone: (479) 575-6584
Office Hour: Tu. Th. 11:00-12:00

**Required Material:**

- Textbook: Luis F. Chaparro, *Signals and Systems Using Matlab*, Academic Press, 2010. ISBN :978-0-12-374716-7
- Software: Matlab

**Reference:**

- S.S. Soliman, M.D. Srinath, *Continuous and discrete signals and systems*, 2nd Ed., Prentice Hall, 1998. (Optional)

**Prerequisites:**

Calculus III, Differential Equation (Co-requisite), Electrical Circuits I or equivalent

- Knowledge of integration, differentiation, and differential equations
- Knowledge of algebra
- Familiar with basic circuit analysis

**Learning Objectives:**

Continuous signals and systems, linear system analysis, convolution, Laplace transform, Fourier series, Fourier transform, discrete-time signals and systems

**Grading:**

- Test 1 26%
- Test 2 26%
- Test 3 26%
- Homework 22%

- A: $90 \leq$ grade $\leq 100$
- B: $80 \leq$ grade $< 90$
- C: $70 \leq$ grade $< 80$
- D: $60 \leq$ grade $< 70$
- F: $0 \leq$ grade $< 60$

- Due dates for homework will be strictly enforced. Late submission within one week after due date will receive a 20% grade deduction, and no credit if submitted after one week from the due date.
- If for some legitimate reason (sickness, death in the family, etc.), you cannot take a **test** on the scheduled day, you must notify the instructor **prior** to the exam.

**Online Resources:**

- Course materials (Slides, Homework, Lecture Notes, Labs, References, etc) can be found at https://wuj.hosted.uark.edu/teaching/eleg3124/eleg387v.html
- Please check course website **at least once per week** for updates.
- Matlab can be downloaded here: https://its.uark.edu/help/ta/matlab-setup.php

**Academic Honesty:**

Each University of Arkansas student is required to be familiar with and abide by the University's 'Academic Integrity Policy' which may be found at http://provost.uark.edu/
Students with questions about how these policies apply to a particular course or assignment should immediately contact their instructor.

**Tentative Schedule:**

- Week 1: Ch.1 Continuous-Time Signals
- Week 1: Ch. 1 Continuous-Time Signals
- Week 2: Ch. 2 Continuous-Time Systems
- Week 2: Ch. 2 Continuous-Time Systems

- Week 3: Ch. 2 Continuous-Time Systems
- Week 3: Ch. 4 Fourier Series (**Test 1, covers Chs. 1 & 2**)
- Week 4: Ch. 4 Fourier Series
- Week 4: Ch. 4 Fourier Series
- Week 5: Ch. 5 Fourier Transform
- Week 5: Ch. 5 Fourier Transform
- Week 6: Ch. 5 Fourier Transform (**Test 2, covers Chs. 4, 5.1-5.3**)
- Week 6: Ch. 3 Laplace Transform
- Week 7: Ch. 3 Laplace Transform
- Week 7: Ch. 3 Laplace Transform
- Week 8: Ch. 3 Laplace Transform
- Week 8: Ch. 6 Discrete-Time System **(Test 3, covers Chs. 5.4, 3, 6)**

**The above schedule is subject to change without prior notice.**